

ITI Cmos Analog IC Design – Cadence Tools

Lab 02 Report — Common Source Amplifier

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Part 1: Sizing Chart

Design specification

Parameter	Value	Notes
Technology	1.8 um	Given
VDD	V_DD = 1.8 V	Given
A_v	7	Randomly generated
L	2 u	Given
I_DQ	60 uA	Randomly generated

1.1 Hand analysis

The output common-mode level is set to $V_{DD}/2$ for maximum output swing:

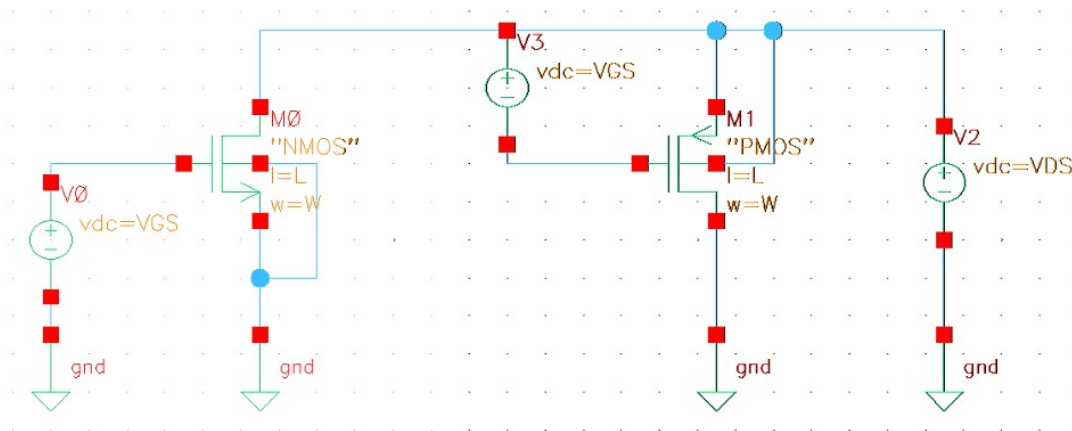
$$V_{RD} = V_{DD} / 2 = 1.8 / 2 = 0.9 \text{ V}$$

$$R_D = V_{RD} / I_{DQ} = 0.9 \text{ V} / 60 \mu\text{A} = 15 \text{ k}\Omega$$

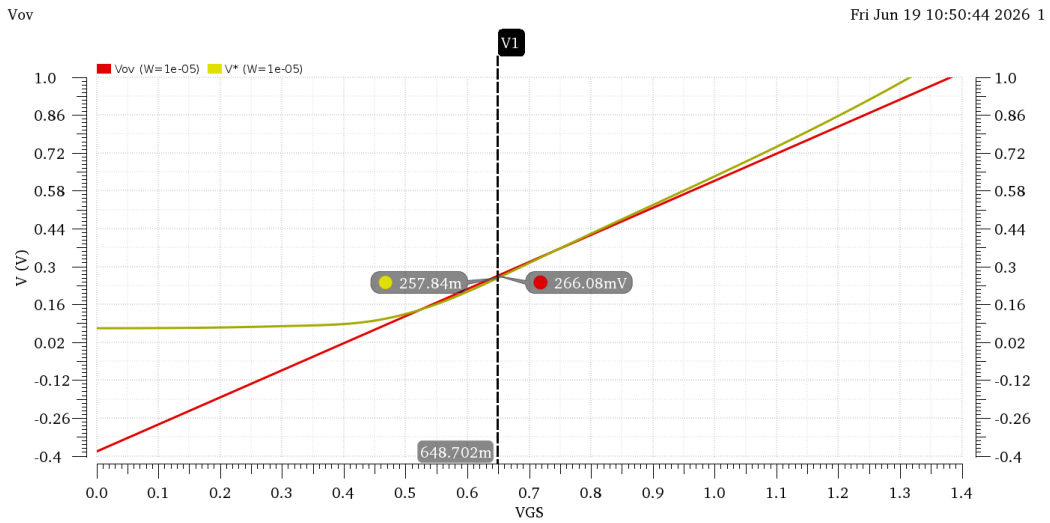
$$\text{From } |A_v| = 2 \cdot V_{RD} / V^*,$$

$$V^*_Q = 2 \cdot V_{RD} / |A_v| = 2(0.9) / 7 = 0.2571 \text{ V} \approx 257.1 \text{ mV}$$

1.2 NMOS Characterization.



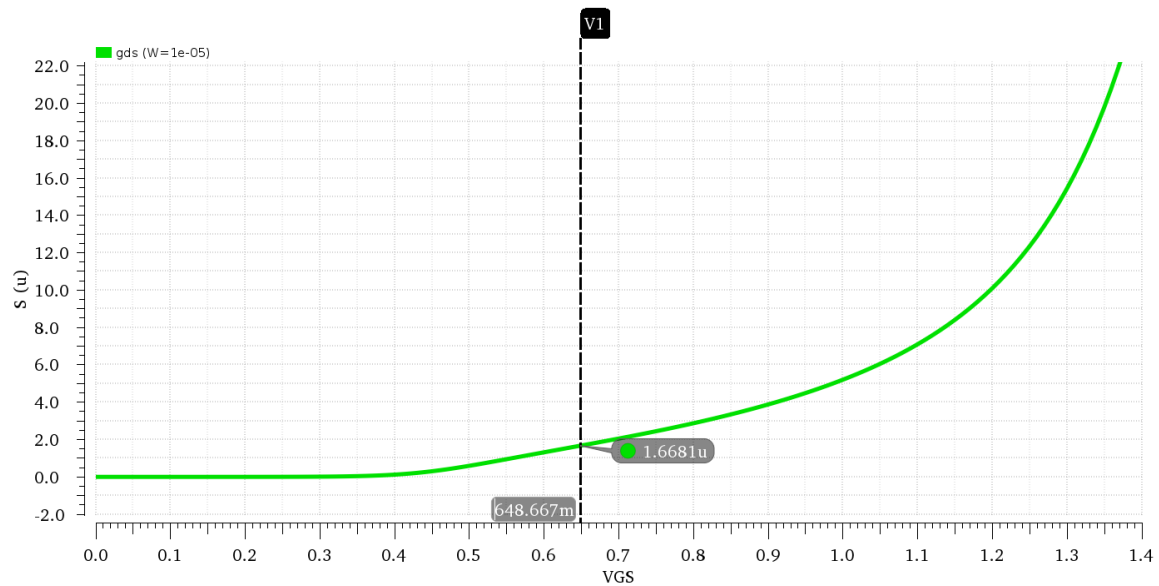
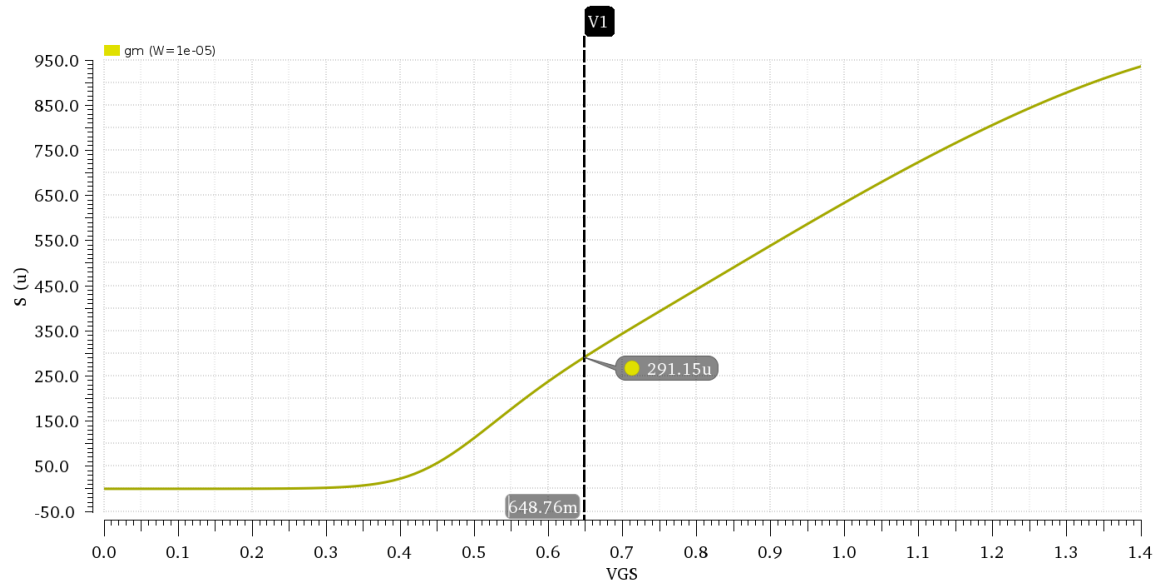
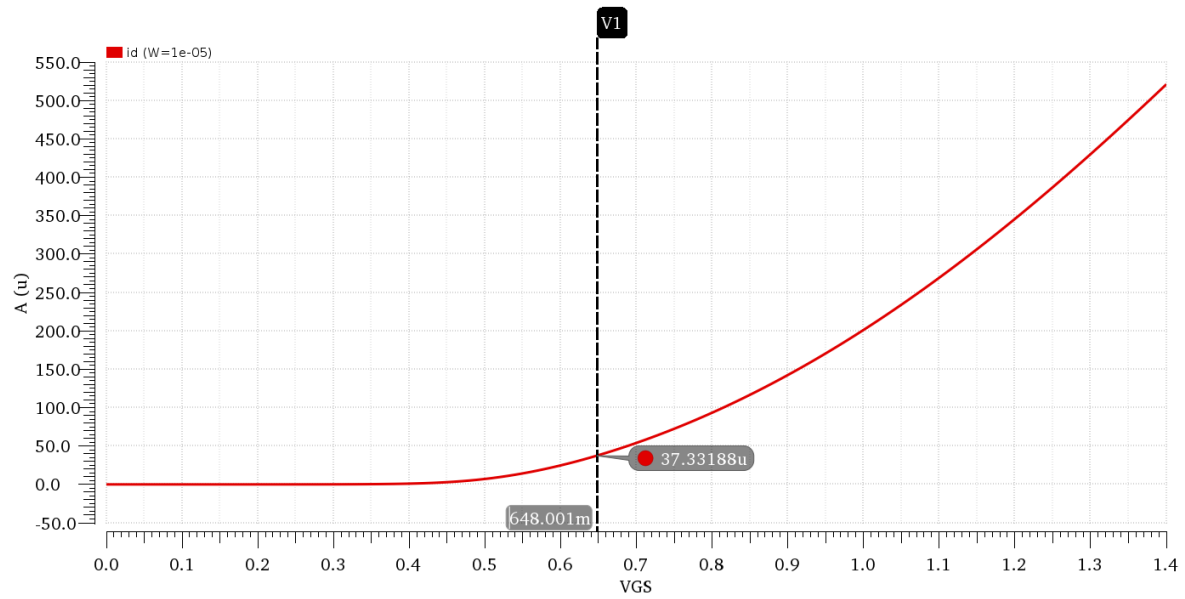
V^* and V_{ov} vs V_{GS}



From the chart

Parameter	Value
V^*_Q (target)	257.1 mV
V_{ovQ}	266.08 mV
V_{GSQ}	257.84 mV

I_{D,g_m} and g_{ds} vs V_{GS}



From chart

Parameter	Value at V_{GSQ} ($W = 10\ \mu\text{m}$)
I_{DX}	37.331 μA
g_{mX}	291.15 μS
g_{dsX}	1.6681 μS

1.3 Scaling to the Design Width and parameters

I_D scales proportionally with W Using ratio and proportion:

W	I_D	g_m	g_{ds} ($1/r_o$)
10 μm	37.331 μA (from chart)	291.15 μS (from chart)	1.6681 μS (from chart)
$W = 60 / 37.331 = \mathbf{16.06\ \mu\text{m}}$	$I_{DQ} = 60\ \mu\text{A}$ (spec)	$g_m = 291\ \mu\text{S} \times 60 / 37.331 = \mathbf{2.7\ \mu\text{S}}$	$G_m = 1.6681\ \mu\text{S} \times (60 / 37.331) = \mathbf{2.7\ \mu\text{S}}$

Check gain

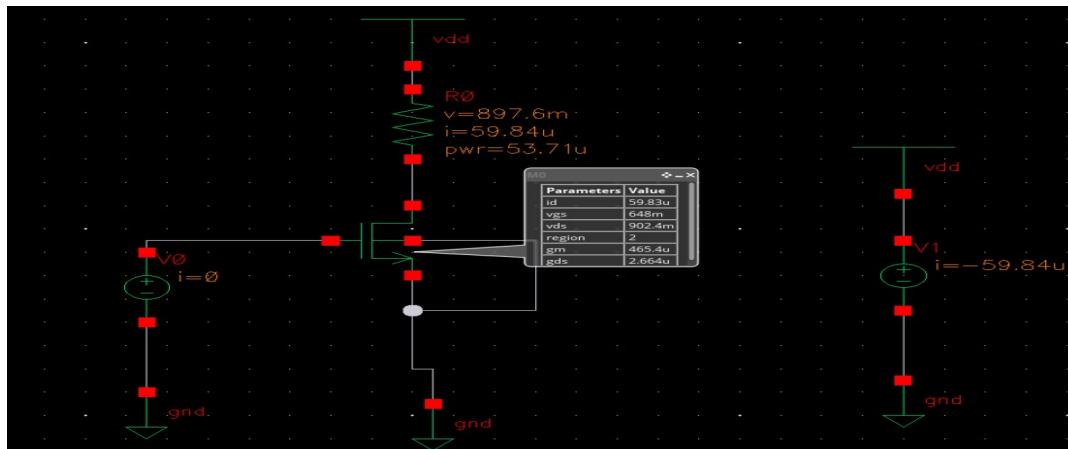
Check that $A_v = -g_m Q(R_D \parallel r_o)$ meets the required gain spec:

$$|A_v| = g_m Q \times (R_D \parallel r_o) = \mathbf{6.8 \text{ (meet specs)}}$$

Part 2 : Common source

Desing specs

Parameter	Value	Notes
Technology	1.8 um	Given
VDD	V_DD = 1.8 V	Given
A_v	7	Randomly generated
L	2 u	Given
I_DQ	60 uA	Randomly generated
W	16 um	From design



1 OP and AC Analysis

DC Operating Point

Parameter	Simulated (OP)	Part 1 (hand calc)
I_D	59.83 uA	60 μ A
V_GS	648 mV	648 mV
V_DS	902.6 mV	0.9 V
Region	saturation (2)	saturation
g_m	465.4 uS	468.4 uS
g_ds	2.664 uS	2.7 us

M0

Parameters	Value
id	59.83u
vgs	648m
vds	902.6m
region	2
gm	465.4u
gds	2.664u

R_o vs. R_D

$$r_o = 1/g_{ds} = 372 \text{ k} \quad R_D = 15 \text{ k}$$

r_o is about 24× larger than R_D, so R_D dominates the parallel combination, so ignoring it will be justified.

At minimum L r_o will decrease more so the parallel combination will increase so error will be increased

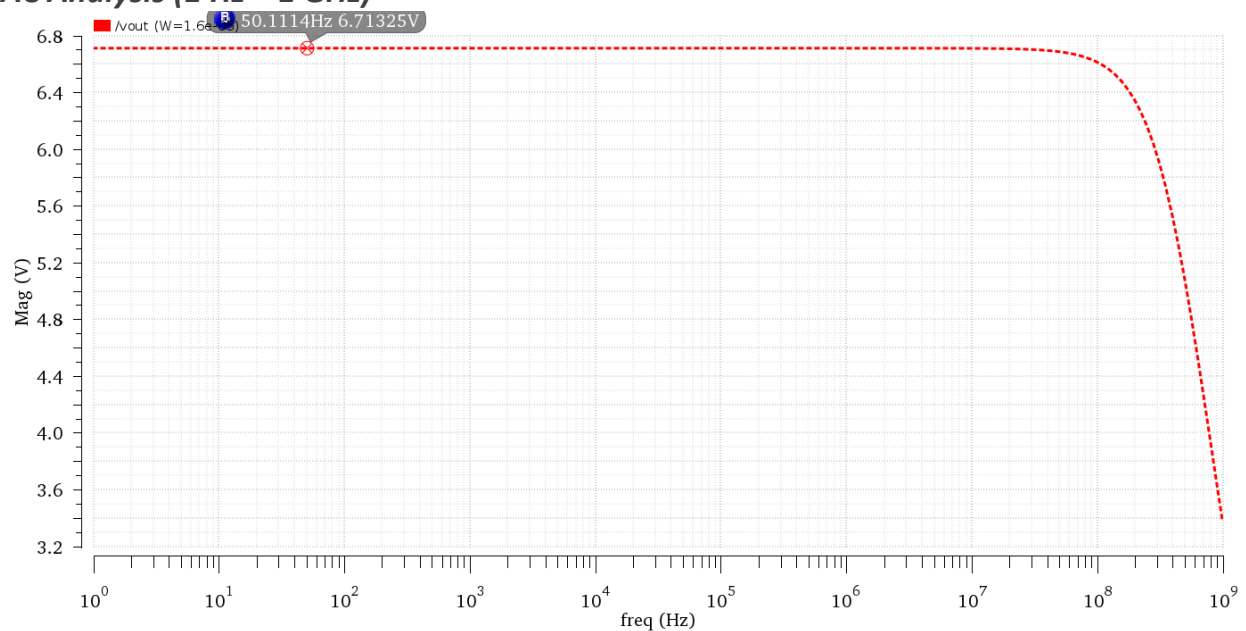
Intrinsic Gain & amplifier gain

$$A_0 = g_m \times r_o = 137$$

$$|A_v| = g_m(R_D \parallel r_o) = 6.75$$

|A_v| ≪ A₀, because of |A_v| = gm(RD||r_o) while A₀ = gm·r_o, and parallel combination of r_D and r_o is much smaller than r_o

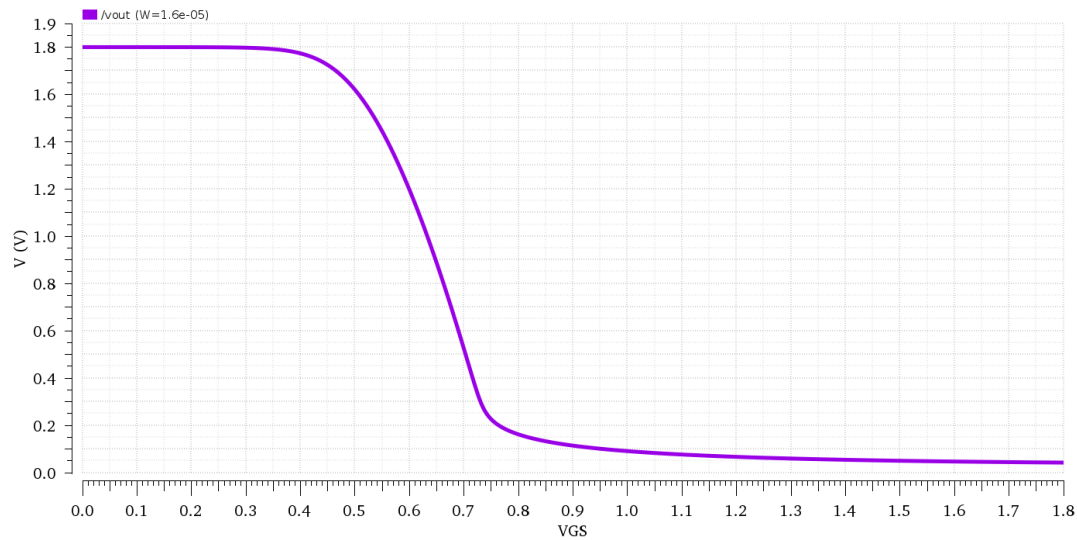
AC Analysis (1 Hz – 1 GHz)



Measured DC gain = 16.58 dB = 6.8 (meet specs)

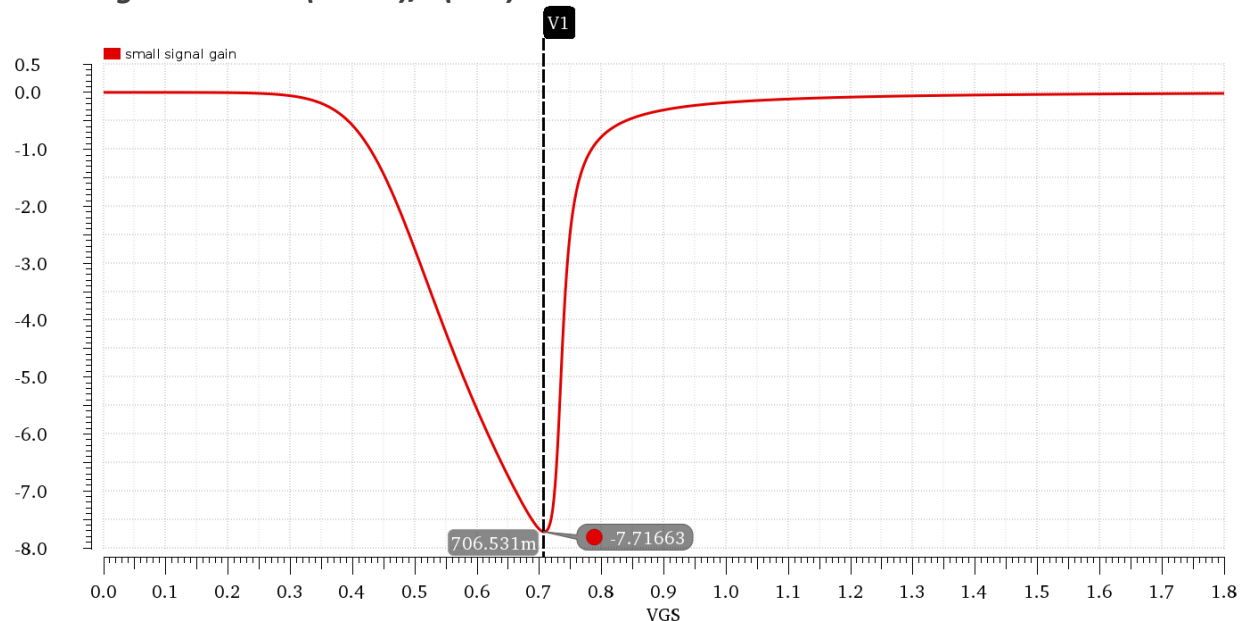
2 Gain Non-Linearity

VOUT vs. VIN (DC Sweep)



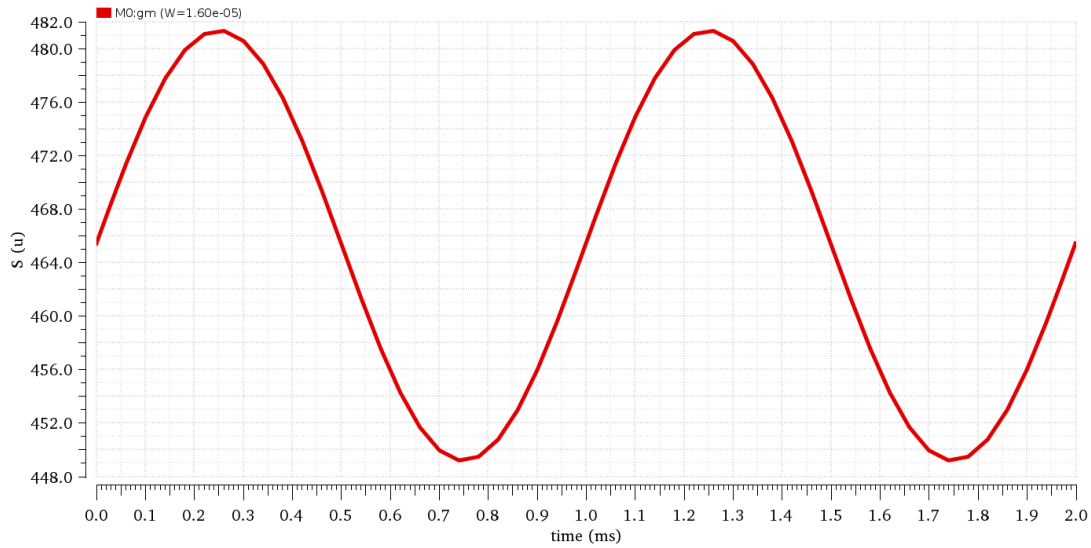
it's not linear because $V_{out} = V_{DD} - I_D \cdot R_D$, and I_D is dependent on V_{IN} (VGS) so when VGS increase i_D increase so V_{out} will decrease.

Small-Signal Gain = $d(VOUT)/d(VIN)$



The small-signal gain is not linear: it is zero at extremes and peaks $V_{IN} \approx V_{GSQ}$, with peak value 7.71663. This confirms the small-signal gain is only valid in a narrow region around the bias point.

Transient g_m vs. Time

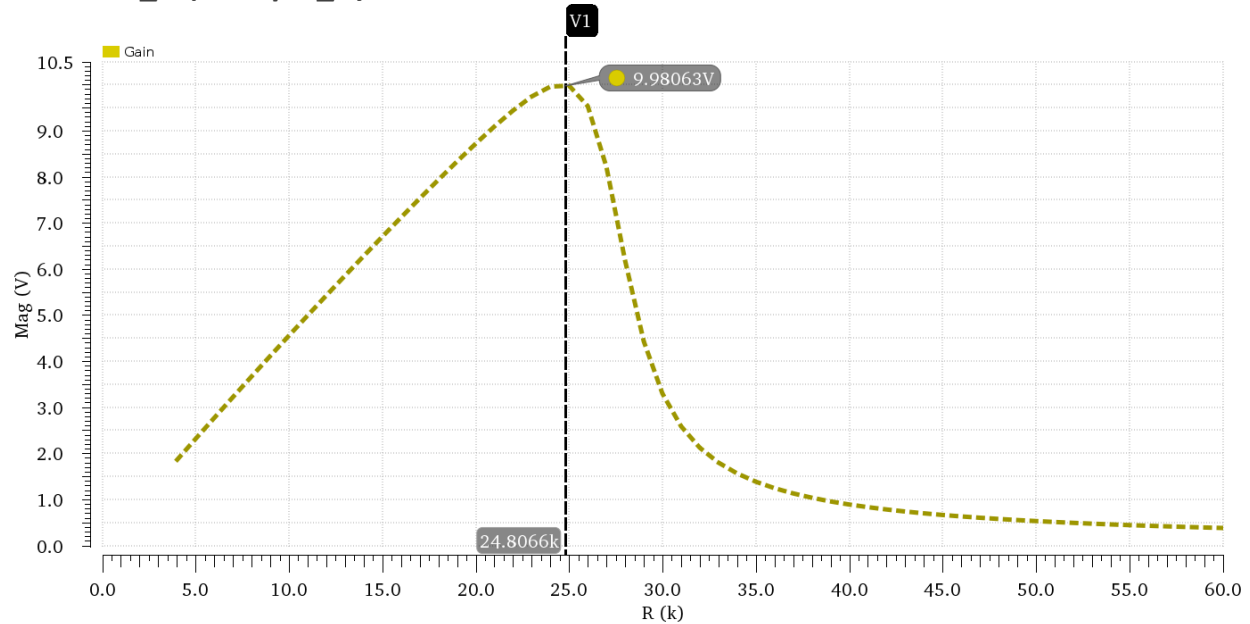


g_m varies with the input, it oscillates at 1kHz, tracking the V_{GS} . This shows the small signal model is only valid for small signals around the bias point only.

The amplifier is linear only for small-signal variations around a fixed bias point. For large input signals, it becomes nonlinear because g_m changes with the instantaneous operating point, causing the gain to vary with the input.

3: Maximum Gain(optional)

Gain vs. R_D (sweep R_D)



Rising part: when $r_d \ll r_o$ the parallel combination is equal to R_D so gain will increase almost linear with R_D .

Falling part: As R_D keeps increasing at fixed I_D , the voltage drop across it ($V_{RD} = I_D \cdot R_D$) keeps growing too but V_{DD} is fixed, so V_{DS} must decrease and device may leave saturation region

R_D at maximum gain = **24.8006 k** Maximum gain = **9.98 V**

Analytical Maximum Gain

Maximum gain will be in Saturation-edge :

$$V_{DD} - I_D \cdot R_D = V^*$$

$$R_{D_max} = (V_{DD} - V^*) / I_D = (1.2 - .2571) / 60\mu = 25.715 \text{ k}\Omega$$

$$|A_v|_{max} \approx g_m(R_{D,max} \parallel r_o) = 11.2$$

Simulation vs. Analysis Comparison

	Analytical	Simulation (from chart)
R_{D_max}	25.17 k Ω	24.8 k Ω
$ A_v _{max}$	11.2	9.9

Available Signal Swing at Max Gain

$$V_{DS,max} = V_{DD} - (I_D \times R_{D_max}) = 1.8 - (60 \mu \times 24.8 \text{ k}) = \mathbf{312 \text{ mV}}$$

scaling down VDD is not good for gain. Since $R_{D_max} = (V_{DD} - V^*)/I_{DQ}$, reducing VDD will decrease R_{D_max} , which lowers the maximum achievable gain.